

UNG FACILITIES PLANNING & GIS

Version current

February 2025

One-time setup tool — generates the building name crosswalk via fuzzy matching

BuildingCrosswalk.py

Purpose

A setup and maintenance tool, not part of the weekly pipeline. Extracts unique building names from parsed AD data, extracts building codes from the GIS BUILDINGCODES domain, and uses fuzzy string matching to suggest mappings between them. Output is an Excel file for manual review. The verified crosswalk it produces is used by every downstream pipeline step.

When to Run

- Once during initial setup — already complete; Building_Crosswalk.xlsx exists and is verified
- When new buildings appear in AD that are not in the crosswalk — identified via AD_Lookup_Diagnostic (see Tool 13)
- When new buildings are added to GIS

Requirements

- ArcGIS Pro / arcpy (reads BUILDINGCODES domain from GDB)
- Parsed AD Excel

Outputs

Item	Value	Notes
Building_Crosswalk.xlsx	ActiveDirectoryParsing\	Three sheets: Building_Crosswalk, GIS_Domain_Reference, Instructions
Verified_GIS_Code column	In workbook	Fill this in manually after reviewing suggestions — this is what the pipeline reads

After Running

Open Building_Crosswalk.xlsx, review the Suggested_GIS_Code column, and fill in Verified_GIS_Code for each row. Focus on high Record_Count buildings first. Buildings marked Manual Review Needed have similarity scores below 0.6 and require careful attention. Save the file — the pipeline reads Verified_GIS_Code on every run.